

Research Plan — Extended Paper

Does the Attention/MLP Division of Labour Scale to Domain Level?

Target: EMNLP 2026 | Prior work: [ICLR 2026 submission](#)

1 The Question

A transformer has two main computational components inside each layer: the **attention sublayer** and the **MLP sublayer**. The mechanistic interpretability literature has established, at the level of individual tokens and specific factual associations, that these two components play different functional roles. Attention handles routing and contextual organisation — moving and aligning information. MLPs function as key-value memories — storing and retrieving associations between input patterns and output content. [This picture has been built up carefully through work on feed-forward layers as memory, factual knowledge localisation, and circuit-level analysis of individual attention heads.](#)

What has not been established is whether this same division holds at a qualitatively coarser level of abstraction: **entire semantic domains**. When a model processes a physics abstract versus an economics abstract, is the same organisational principle at work? Does attention route at the domain level the same way it routes at the token level? Does MLP store domain-specific knowledge the same way it stores individual facts?

The central question: Does the attention/MLP functional division that exists at the token and fact level scale up coherently to domain-level processing — and if so, what does each component actually contribute, and is that contribution localised or distributed across layers?

This is not a restatement of what is already known. The token-level results are all compatible with a weak interpretation: that the attention/MLP division is simply an architectural artifact of how gradients flow through different structures during training. A division that operates at the level of individual tokens does not necessarily organise coherently at the level of semantic domains. Whether it does is an open empirical question, and answering it constrains what kind of system a trained transformer actually is.

2 Why This Question Matters

If the division *does not* scale to domain level — if both attention and MLP contributions are equally and diffusely distributed across all layers when processing domain-specific text — then the token-level functional specialisation is a fine-grained local phenomenon with no higher-level organisational counterpart. The model is, at the domain level, essentially a diffuse function approximator. Mechanistic interpretability tools built on the token-level picture would not extend upward in abstraction.

If the division *does* scale, and moreover if it is **localised** — meaning the routing function is compressed into a small number of identifiable attention layers rather than distributed across all of them — then transformers develop a hierarchical organisational principle. The same architectural

seam (attention vs. MLP) that organises computation at the token level is being used to organise computation at the semantic domain level, and the routing function is not diffuse but structurally concentrated. This changes the foundational picture of what a trained transformer is: not a diffuse statistical machine, but a system with identifiable, reusable structural roles operating across levels of abstraction. That is a meaningful constraint on how we think about interpretation, intervention, and what mechanistic analysis can tell us.

The prior work (ICLR 2026 submission) found evidence for this picture but with three critical weaknesses: the experiments measured component *outputs* rather than component *contributions* to the residual stream, the causal experiment was technically flawed and uninterpretable, and the dataset mixed surface-form-different corpora (code, medical text, finance) where separation could be trivially explained by vocabulary. This work fixes all three.

3 Our Positioning

The existing literature establishes the attention/MLP division at the token and fact level. We ask whether this division scales to domain-level processing, and if so, whether it is organised (localised to specific layers) or diffuse. Our main contribution is to characterise the **structure** of this division — not just that it exists, but where it lives and whether it causally controls domain-level behaviour.

We test this on arXiv abstracts across six academic disciplines: mathematics, physics, computer science, statistics, quantitative biology, and economics. This dataset is the right choice precisely because all six domains share the same surface form — all are formal academic abstracts with the same genre structure. Any domain separation we observe in internal representations is therefore genuinely semantic, not a vocabulary artifact. The cs/statistics boundary is intentionally included as the hardest test case.

We study domain-level processing through a single unifying mathematical object — the **domain concept vector** — which we define rigorously, and then use consistently across all three experiments. Each experiment asks a different question, but all three speak to the same objects and all three results reinforce the same claim. This is not a collection of loosely related analyses; it is a single argument made from three angles.

4 Mathematical Foundations

4.1 The Residual Stream and Component Contributions

In a standard transformer with L layers, the computation at each layer ℓ updates a residual stream $\mathbf{x}^\ell \in \mathbb{R}^d$:

$$\mathbf{x}^{\ell+1} = \mathbf{x}^\ell + \underbrace{\mathbf{A}^\ell(\mathbf{x}^\ell)}_{\Delta \mathbf{a}^\ell} + \underbrace{\mathbf{M}^\ell(\mathbf{x}^\ell + \Delta \mathbf{a}^\ell)}_{\Delta \mathbf{m}^\ell} \quad (1)$$

The quantities $\Delta \mathbf{a}^\ell$ and $\Delta \mathbf{m}^\ell$ are what attention and MLP respectively *add* to the stream at layer ℓ . These are **contributions**, not outputs. This distinction matters critically. Prior work measured $\mathbf{x}^\ell + \Delta \mathbf{a}^\ell$ (the full stream after attention) and called it “the attention output.” But this quantity carries the full residual stream signal — it cannot tell you what attention specifically contributed. We work directly with $\Delta \mathbf{a}^\ell$ and $\Delta \mathbf{m}^\ell$ throughout.

4.2 Domain Concept Vectors

Let $\mathcal{D}_1, \dots, \mathcal{D}_K$ be K domains. For any text x , let $\mathbf{x}^\ell(x)$ denote the residual stream at layer ℓ .

The **global mean** of the residual stream at layer ℓ is:

$$\boldsymbol{\mu}^\ell = \mathbb{E}_x[\mathbf{x}^\ell(x)]$$

This is the centre of mass of all representations — what is common to every text regardless of domain. It encodes generic structure: “this is formal writing,” “this is a sequence of tokens.” It has nothing to do with any specific domain.

The **domain concept vector** for domain k at layer ℓ is:

$$\mathbf{d}_k^\ell = \mathbb{E}_{x \sim \mathcal{D}_k}[\mathbf{x}^\ell(x)] - \boldsymbol{\mu}^\ell \quad (2)$$

Geometrically: you first find the centroid of all representations, move the origin there, and then ask in which direction does domain k sit? The vector $\mathbf{d}_k^\ell$ points from the centre of all text toward the centre of domain k . Its magnitude $\|\mathbf{d}_k^\ell\|$ measures how far domain k sits from the average — how distinct it is. Subtracting $\boldsymbol{\mu}^\ell$ removes shared background signal and isolates what is genuinely domain-specific.

We define analogous **component-specific contribution vectors**:

$$\mathbf{d}_k^{\text{attn},\ell} = \mathbb{E}_{x \sim \mathcal{D}_k}[\Delta \mathbf{a}^\ell(x)] - \mathbb{E}_x[\Delta \mathbf{a}^\ell(x)] \quad (3)$$

$$\mathbf{d}_k^{\text{mlp},\ell} = \mathbb{E}_{x \sim \mathcal{D}_k}[\Delta \mathbf{m}^\ell(x)] - \mathbb{E}_x[\Delta \mathbf{m}^\ell(x)] \quad (4)$$

These three objects satisfy an exact additive decomposition identity:

$$\mathbf{d}_k^{\ell+1} = \mathbf{d}_k^\ell + \mathbf{d}_k^{\text{attn},\ell} + \mathbf{d}_k^{\text{mlp},\ell} \quad (5)$$

This follows directly from the residual stream update rule. **The domain signal in the stream at layer $\ell + 1$ is exactly the domain signal from the previous layer plus what attention added plus what MLP added.** This identity is the mathematical backbone connecting all three experiments.

4.3 The Simplex Test

Park et al. (2024) showed that when a model cleanly separates K categories, the K mean-shifted concept vectors form a **regular simplex** in representation space — they are equally spaced in all directions around the origin, like the vertices of an equilateral triangle (for $K = 3$) generalised to higher dimensions.

For a regular K -simplex, the pairwise cosine similarity between any two vertices equals exactly $-1/(K - 1)$. For $K = 6$ domains, the ideal value is $-1/5 = -0.2$.

We compute the 6×6 pairwise cosine similarity matrix of domain concept vectors $\{\mathbf{d}_k^\ell\}_{k=1}^6$ at every layer, for the residual stream and for each component’s contributions separately. The **simplex deviation** at layer ℓ for a component is the Frobenius distance between the measured cosine similarity matrix and the ideal simplex matrix (all off-diagonals equal to -0.2 , all diagonals equal to 1).

If attention contributions form near-simplex geometry (low simplex deviation) at hotspot layers while MLP contributions do not, this means attention is organising the six domains into a clean, symmetric, maximally-separated geometric structure — exactly what a routing mechanism should do. MLP doing something geometrically different (possibly more hierarchical, or more entangled) is consistent with the storage and execution role.

5 Dataset

Domain	arXiv Categories	Why included
Mathematics	<code>math.*</code>	Heavily symbolic, distinct from natural language prose
Physics	<code>hep-th</code> , <code>cond-mat</code> , <code>quant-ph</code>	Close to mathematics but experimentally grounded
Computer Science	<code>cs.CL</code> , <code>cs.CV</code> , <code>cs.LG</code>	Empirical and systems-focused
Statistics	<code>stat.ML</code> , <code>stat.ME</code> , <code>stat.TH</code>	Methodologically close to CS; hardest boundary
Quant. Biology	<code>q-bio.NC</code> , <code>q-bio.GN</code>	Biological data and systems
Economics	<code>econ.EM</code> , <code>econ.TH</code>	Social science reasoning and modelling

Table 1: Six arXiv domains. The cs/statistics pair is the intentionally hard case.

- 5,000 abstracts per domain (balanced), 30,000 total.
- Filter cross-listed papers — keep only single-primary-category papers.
- Truncate all abstracts to 256 tokens for comparability.
- Split: 70/15/15 train/val/test, stratified by domain. Fix the split before any experiment.
- Model: **Llama-3.1-8B** (base, not instruction-tuned). One primary model.

Why arXiv? All six domains share the same genre: formal academic abstracts. The writing structure is identical — problem statement, method, results. Any domain separation we observe in internal representations is therefore genuinely about semantic content, not about vocabulary distribution or writing style. This is a harder and more honest test than the prior work’s mixed corpus of code, medical text, and finance.

6 Experiment 1: Domain Concept Vectors and Geometric Structure

Question: Do the model’s representations organise the six domains into a coherent geometric structure? And does this structure look different for attention contributions versus MLP contributions?

This experiment establishes the foundational objects — the domain concept vectors — that everything else is built on.

6.1 What We Compute

For every layer $\ell \in \{1, \dots, L\}$ and every domain $k \in \{1, \dots, 6\}$, we extract the following:

- $\mathbf{x}^\ell(x)$ — the full residual stream at layer ℓ for text x , extracted at the last token position.
- $\Delta \mathbf{a}^\ell(x)$ — the attention contribution at layer ℓ : the output of the attention sublayer before being added to the stream. Concretely, this is the output of `o_proj` before the residual add.
- $\Delta \mathbf{m}^\ell(x)$ — the MLP contribution at layer ℓ : the output of the `down_proj` before the residual add.

We then compute domain concept vectors $\mathbf{d}_k^\ell$, $\mathbf{d}_k^{\text{attn}, \ell}$, and $\mathbf{d}_k^{\text{mlp}, \ell}$ for all k and ℓ as defined in Section 4.

6.2 Separability: Fisher Discriminant Ratio

The **multiclass Fisher Discriminant Ratio** at layer ℓ is:

$$\text{FDR}^\ell = \frac{\text{tr}(S_B^\ell)}{\text{tr}(S_W^\ell)}$$

where the between-class scatter is:

$$S_B^\ell = \sum_{k=1}^K n_k \mathbf{d}_k^\ell (\mathbf{d}_k^\ell)^\top \quad \Rightarrow \quad \text{tr}(S_B^\ell) = \sum_{k=1}^K n_k \|\mathbf{d}_k^\ell\|^2$$

and the within-class scatter is:

$$S_W^\ell = \sum_{k=1}^K \sum_{x \in \mathcal{D}_k} (\mathbf{x}^\ell(x) - \boldsymbol{\mu}_k^\ell) (\mathbf{x}^\ell(x) - \boldsymbol{\mu}_k^\ell)^\top$$

Note the key identity: $\text{tr}(S_B^\ell) = \sum_k n_k \|\mathbf{d}_k^\ell\|^2$. **The between-class scatter is literally the sum of squared norms of the domain concept vectors.** The separability metric is a direct function of the concept vectors — there is no need to define separability separately, it falls out of the geometry automatically.

We compute FDR separately on:

1. The full residual stream $\mathbf{x}^\ell$
2. Attention contributions $\Delta \mathbf{a}^\ell$
3. MLP contributions $\Delta \mathbf{m}^\ell$

Plot all three curves across layers on a single figure. The **localisation claim** is: the attention contribution FDR curve has sharp peaks (high variance across depth), while the MLP contribution FDR curve is flatter.

We quantify localisation formally using the **coefficient of variation** ($\text{CV} = \text{std}/\text{mean}$) of each layer-wise FDR curve. High CV means the signal is compressed into a few layers. Low CV means it is distributed uniformly. This is a single interpretable number per component.

6.3 The Vocabulary Baseline

Early layers may show high separability simply because different domains use different vocabulary. This would not be an interesting finding. We establish a **token distribution baseline**: compute the FDR on raw unigram token distributions per domain (i.e., the embedding-weighted mean with no forward pass). Any layer whose FDR does not exceed this baseline is showing separability that is purely vocabulary-driven. We report which layers exceed the baseline, and only those layers are considered meaningful.

6.4 Simplex Test

At each layer, compute the 6×6 pairwise cosine similarity matrix of $\{\mathbf{d}_k^\ell\}$. Compute the **simplex deviation**: Frobenius distance between this matrix and the ideal simplex matrix (off-diagonals = -0.2 , diagonals = 1). Do the same for $\{\mathbf{d}_k^{\text{attn},\ell}\}$ and $\{\mathbf{d}_k^{\text{mlp},\ell}\}$.

If attention contributions approach simplex geometry at the same layers where FDR peaks, this is strong geometric evidence of a routing mechanism: the model is organising domain representations into clean, symmetric, maximally-separated directions specifically through what attention adds at those layers.

6.5 Visualisation

At the layer with the highest attention contribution FDR, produce a UMAP projection of all 30,000 residual stream points coloured by domain. Also produce the same UMAP for attention contributions and MLP contributions at that layer. These are not the primary evidence but are useful for readers to develop geometric intuition.

Expected result: Attention contribution FDR peaks sharply at 2–4 specific mid-depth layers. MLP contribution FDR is flatter and lower at those same layers. The attention contribution simplex deviation is lower (closer to ideal geometry) at the peak layers. UMAP shows clean clusters for attention contributions, messier clusters for MLP contributions. Critically: the cs/statistics boundary is the weakest — we report it separately. If the pattern holds even there, that strengthens the claim.

7 Experiment 2: Fine-Tuning Analysis

Question: When you fine-tune the model on a specific domain, where do the weight updates go? Are the MLP updates larger, and more importantly, are they *domain-specific* rather than just generically large?

This experiment tests the storage hypothesis from the adaptation side. The key improvement over the prior work: we do not just measure *how much* weights change. We measure whether MLP weight changes are *aligned with the domain concept vectors* — i.e., whether the updates specifically modify the directions in weight space that carry domain identity.

7.1 Setup

Fine-tune six separate models, one per domain, from the same Llama-3.1-8B base checkpoint using LoRA (rank 16, $\alpha = 32$, applied to $W_q, W_v, W_{\text{up}}, W_{\text{down}}$ at all layers). Three epochs on each domain’s 3,500 training abstracts.

This gives six domain-specialised models. For each, compute the effective weight update at each layer. For LoRA, the update to a weight matrix W is $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times d}$. We work with ΔW directly.

7.2 Metric 1 — Weight Change Norms (Replication)

The relative weight change:

$$\delta_{\text{mlp}}^{(k),\ell} = \frac{\|\Delta W_{\text{mlp}}^{(k),\ell}\|_F}{\|W_{\text{mlp,base}}^\ell\|_F}, \quad \delta_{\text{attn}}^{(k),\ell} = \frac{\|\Delta W_{\text{attn}}^{(k),\ell}\|_F}{\|W_{\text{attn,base}}^\ell\|_F}$$

Plot layer-wise δ_{attn} vs δ_{mlp} for each of the 6 domains. This replicates the prior work’s Figure 2, now on arXiv. The 2–3 $\times$ difference should hold across domains. If it does, it supports the claim and shows generality. If it does not, that is a significant finding in itself.

7.3 Metric 2 — Domain-Specificity of Updates

This is the key new metric. For each layer ℓ , we have six weight update matrices: $\Delta W_\ell^{(1)}, \dots, \Delta W_\ell^{(6)}$ (one per domain). We ask: are these updates differentiated from each other?

Compute the pairwise cosine similarity between vectorised weight updates:

$$\text{sim}(k, k')^\ell = \frac{\langle \text{vec}(\Delta W_\ell^{(k)}), \text{vec}(\Delta W_\ell^{(k')}) \rangle}{\|\text{vec}(\Delta W_\ell^{(k)})\| \cdot \|\text{vec}(\Delta W_\ell^{(k')})\|}$$

Do this separately for MLP and attention updates. Produce 6×6 cross-domain update similarity matrices at selected layers.

The prediction: MLP updates should be more domain-differentiated (lower cross-domain cosine similarity) than attention updates. This means when you fine-tune on physics, the MLP update looks different from when you fine-tune on economics — the updates are genuinely learning domain-specific things. Attention updates, by contrast, should be more similar across domains — attention adapts generically regardless of which domain you are learning.

This directly answers reviewer R4WC’s complaint about the prior work: that large MLP updates might just mean MLP has a favourable optimisation landscape. Domain-differentiation of updates is not explained by optimisation dynamics — it is a content argument.

7.4 Metric 3 — Alignment with Domain Concept Vectors

The deepest metric. For each domain k and each layer ℓ , we ask: does the MLP weight update $\Delta W_{\text{mlp}}^{(k),\ell}$ operate specifically in the domain- k direction in representation space?

Define the **domain alignment of the weight update**:

$$\rho_k^{\text{mlp},\ell} = \frac{\|\Delta W_{\text{mlp}}^{(k),\ell} \hat{\mathbf{d}}_k^{\text{mlp},\ell}\|}{\|\Delta W_{\text{mlp}}^{(k),\ell}\|_F}$$

where $\hat{\mathbf{d}}_k^{\text{mlp},\ell}$ is the unit domain concept vector for domain k (from Experiment 1). This quantity measures: when you apply the weight update to the domain- k direction in activation space, how large is the output relative to the total norm of the update? High ρ means the update is specifically

moving the domain- k direction. Low ρ means the update is orthogonal to the domain concept vector — it is changing something unrelated to domain identity.

Compute $\rho_k^{\text{mlp},\ell}$ and $\rho_k^{\text{attn},\ell}$ for all domains and layers. The prediction: MLP updates have higher alignment with domain concept vectors than attention updates. This is the quantitative statement that *MLP adapts in the directions that carry domain identity*.

This metric directly uses the concept vectors from Experiment 1 inside the adaptation analysis. The domain concept vector is the glue: it was computed from the forward pass geometry in Experiment 1, and here it is used to evaluate whether training updates specifically modify those geometric directions.

Expected result: MLP updates are 2–3 $\times$ larger in norm (replication), more domain-differentiated (lower cross-domain similarity), and more aligned with domain concept vectors. Attention updates are smaller, more uniform across domains, and less aligned. Pattern holds for all six domains. If any domain breaks the pattern, investigate and report it — the cs/statistics boundary is the most likely candidate.

8 Experiment 3: Causal Activation Patching

Question: Is the domain signal in attention causally responsible for domain-directed behaviour? And if so, does it propagate through MLP layers downstream, or does it ride passively through the stream?

The first two experiments are observational and adaptive. This experiment is causal. It asks: if you surgically change the domain routing signal in attention at a specific layer, does the model’s subsequent computation change in a directed, domain-consistent way?

8.1 Concept Direction Patching

Previous approaches to activation patching replaced the full activation at a given layer with that from a different input. This is too coarse: it transfers not just the domain signal but all content-specific structure in the activation. The results are uninterpretable.

We patch only along the domain concept direction. For a source text $x_A \in \mathcal{D}_A$ and target domain B , the patched attention contribution at layer ℓ^* is:

$$\widetilde{\Delta \mathbf{a}}^{\ell^*}(x_A) = \Delta \mathbf{a}^{\ell^*}(x_A) - \underbrace{\left\langle \Delta \mathbf{a}^{\ell^*}(x_A), \hat{\mathbf{d}}_A^{\text{attn},\ell^*} \right\rangle \hat{\mathbf{d}}_A^{\text{attn},\ell^*}}_{\text{remove domain-}A \text{ component}} + \underbrace{\alpha \cdot \hat{\mathbf{d}}_B^{\text{attn},\ell^*}}_{\text{inject domain-}B \text{ component}} \quad (6)$$

where $\hat{\mathbf{d}}_k^{\text{attn},\ell}$ is the unit domain concept vector for domain k , and α is chosen to match the typical projection magnitude of domain- B representations onto their own concept vector (so the injected signal is calibrated, not arbitrary).

This operation subtracts the domain- A component from the attention contribution and replaces it with a domain- B component of natural magnitude. Everything else in the activation — all the token-level and content-specific structure — is left intact. The patched forward pass then continues normally for all layers after ℓ^* .

8.2 Measuring the Effect

We measure two things:

(1) **Final domain shift.** At the final residual stream $\tilde{\mathbf{x}}^L$, compute:

$$\text{shift}(\ell^*) = \left\langle \tilde{\mathbf{x}}^L - \mathbf{x}^L(x_A), \frac{\mathbf{d}_B^L - \mathbf{d}_A^L}{\|\mathbf{d}_B^L - \mathbf{d}_A^L\|} \right\rangle$$

This measures how much the final representation moved in the direction from domain A toward domain B as a result of the patch. Average across 200 pairs (x_A, x_B) for each of the 15 domain pairs, and across all choices of ℓ^* (one per layer). Plot the resulting curve: domain shift as a function of patch layer ℓ^* . A sharp peak at the same layers where attention contribution FDR peaked in Experiment 1 is the causal confirmation of the localization claim.

Run the same patch on MLP contributions (replacing the domain- A component of $\Delta \mathbf{m}^{\ell^*}$ with domain- B). The prediction: MLP patches produce less directed shift (lower or noisier shift scores) — not because MLP is unimportant, but because the domain routing signal does not live there.

(2) **Propagation analysis.** This is the piece that connects the causal result back to the functional roles.

After a patch at hotspot layer ℓ^* , track the domain shift score at every intermediate layer $\ell > \ell^*$:

$$s^\ell = \left\langle \tilde{\mathbf{x}}^\ell - \mathbf{x}^\ell(x_A), \frac{\mathbf{d}_B^\ell - \mathbf{d}_A^\ell}{\|\mathbf{d}_B^\ell - \mathbf{d}_A^\ell\|} \right\rangle \quad (7)$$

Note that the concept vectors in the denominator are layer-specific and computed from clean forward passes, so they are different at each layer. This makes the projection geometrically well-defined at every layer.

The domain shift score s^ℓ tells you: at layer ℓ , how much has the patched stream moved toward domain B relative to the clean stream? By tracking this across layers, you get a picture of how the injected signal propagates through subsequent computation.

8.3 What Propagation Tells You

There are three possible outcomes, all interpretable:

Signal amplifies through downstream MLP layers. The shift score s^ℓ grows specifically at the MLP sublayers immediately following the patch. This is the clearest confirmation of the routing/execution picture: attention at ℓ^* injected a domain signal, and downstream MLP layers responded to it by amplifying domain- B -consistent computation. The routing decision at attention triggers execution in MLP.

Signal persists but does not amplify. s^ℓ stays roughly constant after the patch. The domain- B signal rides along in the stream without anything actively responding to it. This is a weaker version of the routing hypothesis: attention controls what is in the stream, but MLP does not specially act on domain signals at this level of abstraction. Still interpretable and publishable.

Signal decays. s^ℓ decreases after the patch — downstream layers actively pull the stream back toward domain- A processing. This would mean domain identity is maintained redundantly across multiple layers, not localised at the hotspots. This would challenge the localisation claim and is the most important failure mode to watch for.

The key figure: x-axis is layer index from ℓ^* onward, y-axis is mean shift score averaged across domain pairs, with one curve for patches at hotspot attention layers and one for patches at non-hotspot layers (and MLP). If the hotspot curve rises and the others do not, you have both causal confirmation and a mechanistic account of how the signal propagates.

On circularity. The patch uses concept vectors $\mathbf{d}_k^{\text{attn}, \ell^*}$ (from attention contributions). The measurement uses residual stream vectors $\mathbf{d}_k^\ell$ (from the full stream at each layer). These are different objects computed from different quantities. Patching along the attention contribution direction does not guarantee the residual stream will align with its own domain concept vectors downstream — that must be demonstrated empirically. There is no circularity. **One thing to verify:** the magnitude of the injected perturbation $\|\tilde{\mathbf{x}}^{\ell^*} - \mathbf{x}^{\ell^*}(x_A)\|$ should be small relative to the within-domain spread $\sqrt{\text{tr}(S_W^{\ell^*})}$. Report this ratio to confirm the linear approximation is valid.

9 What the Experiments Say Together

Each experiment asks a different question, all about the same objects:

1. **Concept vectors + separability:** Do attention contributions at specific layers carry disproportionate domain signal, and is it geometrically organised? (*Observational*)
2. **Fine-tuning:** Do weight updates during domain-specific training modify the directions in MLP weight space that correspond to domain concept vectors? (*Adaptive*)
3. **Causal patching:** Does surgically injecting a domain concept direction into attention at hotspot layers cause directed domain shifts that propagate through downstream MLP? (*Causal*)

If all three work, the argument is:

- Attention contributes disproportionate, localised, geometrically-organised domain signal at specific layers (Experiment 1).
- MLP weight updates are larger, more domain-specific, and specifically aligned with domain concept vectors (Experiment 2).
- Patching the domain direction in attention at those specific layers causes directed domain shifts that are amplified by downstream MLP layers (Experiment 3).

Together: the attention/MLP functional division scales to domain level. Routing is compressed into identifiable attention layers. Execution happens in MLP. The organisation is not diffuse but structural.

10 Open Questions Before Starting

1. **Run a sanity check first.** Before the full dataset, run Experiment 1 on 500 examples per domain with one layer. If the concept vectors are reasonable and FDR is non-trivial, proceed. If not, investigate the hooking methodology.

2. **cs/statistics boundary.** Before fixing the corpus, check whether FDR between cs and stat is above the vocabulary baseline at any layer. If not, consider merging them and adding a different sixth domain, or keep them and frame the result as showing the limits of the model’s domain organisation.
3. **Token position.** We use the last-token representation throughout. Also check mean-pooled representations. If results differ substantially, report both and discuss.
4. **Patch calibration.** For Experiment 3, the scale of α in equation 6 needs to be set carefully. Use $\alpha = \mathbb{E}_{x \sim \mathcal{D}_B} [\langle \Delta \mathbf{a}^{\ell^*}(x), \hat{\mathbf{d}}_B^{\text{attn}, \ell^*} \rangle]$ — the average projection of domain- B texts onto their own concept vector. This calibrates the injected signal to match what a real domain- B representation would contribute.
5. **Statistical testing.** Report confidence intervals from bootstrap resampling across abstract pairs for the shift scores in Experiment 3. Report standard deviations across domains for metrics in Experiment 2.